

Suggested Solutions to Exercises 7

1. The assumptions behind the derivation of the Black-Scholes formula are:
 - 1) The stock price follows a geometric Brownian motion
 - 2) We can have any long or short position in the stock
 - 3) There are no transaction costs
 - 4) No dividends
 - 5) No riskless arbitrage
 - 6) Trading is continuous in time
 - 7) Risk-free interest rate is constant and the same for all maturities
2. We have seen that

$$u_i = (\log S_i - \log S_{i-1}) \sim N \left[\left(\mu - \frac{\sigma^2}{2} \right) \tau, \sigma^2 \tau \right]$$

where $\tau = 1/252$ (1 day out of 252 trading days a year). Therefore, if we take logs of the stock prices given in the text, and then calculate the differences, we can find the *daily* sample mean and sample variance of the differences. They will be estimates for $\left(\mu - \frac{\sigma^2}{2} \right) \tau$ and $\sigma^2 \tau$, respectively. We have the following u_i :

$$\begin{aligned} &-0.01265, -0.00104, -0.00486, 0.01006, 0.03353 \\ &-0.01444, -0.00803, 0.00218, 0.00326, 0.00505 \end{aligned}$$

which give

$$\begin{aligned} \bar{x} = 0.00131 &\Rightarrow \text{yearly mean} = 252\bar{x} = 0.3291 \\ s^2 = 0.00019 &\Rightarrow \text{yearly variance} = 252s^2 = 0.0478 \end{aligned}$$

From this we obtain:

$$\hat{\sigma} = \sqrt{0.0478} = 0.21862 = 21.86\%$$

$$\hat{\mu} = 0.32911 + \frac{1}{2} \times 0.0478 = 0.353$$

Now we can find a confidence interval for the next days closing price, S_{11} , from the fact that $\log S_{11} - \log S_{10} \sim N[(\mu - \sigma^2/2)\tau, \sigma^2\tau]$, using the estimates above for the two parameters. A 99% confidence interval for the closing price is obtained from:

$$P \left[-2.57 < \frac{\log S_{11} - \log S_{10} - 0.00131}{\sqrt{0.00019}} < 2.57 \right] = 0.99$$

Substituting in $S_{10} = 137.838$ we obtain

$$P [4.8920 < \log S_{11} < 4.9628] = 0.99$$

which gives a 99% confidence interval for S_{11} equal to $[133.215, 142.9956]$. The price the following day was in fact 134.55.

3. In this case $S_0 = 30$, $X = 29$, $r = 0.05$, $\sigma = 0.25$ and $T = 0.3333$. We can compute the constants d_1 and d_2 as:

$$d_1 = \frac{\log(\frac{30}{29}) + \left(0.05 + \frac{0.25^2}{2}\right) \times 0.3333}{0.25\sqrt{0.3333}} = 0.4225$$

$$d_2 = 0.4225 - 0.25\sqrt{0.3333} = 0.2782$$

Moreover we have $N(0.4225) = 0.6637$ and $N(0.2782) = 0.6096$, and therefore $N(-0.4225) = 0.3363$ and $N(-0.2782) = 0.3904$. Then

- (a) The European call price is

$$30 \times 0.6637 - 29e^{-0.05 \times 0.3333} \times 0.6096 = 2.52$$

- (b) The American call price is the same as the European call price, i.e. £2.52.

- (c) The European put price is

$$29e^{-0.05 \times 0.3333} \times 0.3904 - 30 \times 0.3363 = 1.05$$

- (d) Put-call parity states that $\text{Call} - \text{Put} = S_0 - Xe^{-rT}$. In this case $\text{Call} = 2.52$, $\text{Put} = 1.05$, $S_0 = 30$, $X = 29$ and $e^{-rT} = 0.9835$, and it is easy to verify that the relationship is satisfied.

4. The Black-Scholes formula for a put is

$$P = Xe^{-rT}N(-d_2) - SN(-d_1)$$

so that

$$\begin{aligned} P + S &= Xe^{-rT}N(-d_2) - SN(-d_1) + S \\ &= Xe^{-rT}N(-d_2) + S(1 - N(-d_1)). \end{aligned}$$

We know that $N(x) = 1 - N(-x)$ by the properties of the standard normal distribution, so we have

$$\begin{aligned} P + S &= Xe^{-rT}(1 - N(d_2)) + SN(d_1) \\ &= SN(d_1) - Xe^{-rT}N(d_2) + Xe^{-rT} \\ &= C + Xe^{-rT} \end{aligned}$$

as the price of a call is

$$C = SN(d_1) - Xe^{-rT}N(d_2).$$

5. Use put-call parity

$$p + S = c + Xe^{-r(T-t)}$$

so that

$$p = c - S + Xe^{-r(T-t)}.$$

Here $c = 6$, $S = 23$, $X = 25$, $r = 0.07$ and $T - t = 0.25$ (3 months to expiry = 3/12 years) which gives p , the value of the put option, as £7.57.

6. The implied volatility is the volatility that, when used in the Black-Scholes formula, gives the current market price for the option. Calculate the call option price using the trader's volatility of 15% in the usual Black-Scholes formula. This gives a call option value of 1.69. But the option is trading at 2.00. As a call option price is a strictly increasing function of volatility, the implied volatility must therefore be more than the 15% used by the trader.